

Experiment:3: Use Flexbox/Grid to layout a multi-column webpage.

 **Aim:** To design a multi-column webpage using CSS Flexbox/Grid for structured and responsive layout.

Theory

Flexbox (Flexible Box Layout): A one-dimensional layout system in CSS that arranges items in rows or columns. It is useful for aligning and distributing space among items.

- **CSS Grid Layout:** A two-dimensional layout system that allows developers to create complex layouts with rows and columns.
- **Multi-Column Webpage:** A webpage divided into multiple sections (like header, sidebar, content, footer) arranged neatly using Flexbox or Grid.

Flexbox is best for **row/column alignment**, while Grid is ideal for **overall page structure**.

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Multi-Column Layout</title>
  <style>
    body {
      margin: 0;
      font-family: Arial, sans-serif;
    }
    .grid-container {
      display: grid;
      grid-template-areas:
        "header header"
        "nav content"
        "footer footer";
      grid-template-columns: 200px 1fr;
      grid-template-rows: 80px auto 60px;
      height: 100vh;
    }
    header {
      grid-area: header;
      background: #333;
      color: white;
      text-align: center;
```

```

        padding: 20px;
    }
    nav {
        grid-area: nav;
        background: #f4f4f4;
        padding: 20px;
    }
    main {
        grid-area: content;
        padding: 20px;
        background: #fff;
    }
    footer {
        grid-area: footer;
        background: #333;
        color: white;
        text-align: center;
        padding: 15px;
    }
}
</style>
</head>
<body>

<div class="grid-container">
    <header>
        <h1>My Multi-Column Webpage</h1>
    </header>
    <nav>
        <h2>Navigation</h2>
        <ul>
            <li>Home</li>
            <li>About</li>
            <li>Services</li>
            <li>Contact</li>
        </ul>
    </nav>
    <main>
        <h2>Content Area</h2>
        <p>This is the main content section of the webpage. It is styled using CSS Grid for a neat multi-column layout.</p>
    </main>
    <footer>
        <p>&copy; 2026 My Website</p>
    </footer>
</div>

</body>
</html>

```

✓ Sample Output:

HEADER	
NAVIGATION	CONTENT
Home	This is the main content
About	section styled with Grid.
Services	
Contact	
FOOTER	